

Foreword

The Standard Model—the temple of contemporary particle physics—marks the pinnacle of humanity’s rational exploration with its unparalleled predictive accuracy. Yet, when we examine the very foundations of this temple, we discover a series of unsettling fissures: the core elements that underpin the entire theory—the structure of gauge groups, the absolute nature of chirality, the quantization of charge, and even the patterns of elementary particle mixing—lack any first-principles explanation. The Standard Model can only treat them as “magic numbers” that must be input from experiments, leaving its theoretical foundations open.

The table below reveals the five most profound unsolved mysteries of the Standard Model, along with how **KnotSim** (discrete knot theory) provides a closed-form solution rooted in the rigidity of geometry and topology:

Unsolved Mysteries of the Standard Model	KnotSim's Geometric Enclosure Method	Predictions/Verification
1. The Mystery of the Standard Group Why $SU(3)_C \times SU(2)_L \times U(1)_Y$? Why not some other group?	Topological classification in three-dimensional space (Thurston geometrization) $\rightarrow$ Deriving icosahedral symmetry from ADE classification Breach chains: $SO(10) \rightarrow SU(3) \times SU(2) \times U(1)$	The standardization of group structure is an inevitable consequence of spatial topology, ruling out other possibilities for groups.
2. The Mystery of Chirality Why does the weak force couple only to left-handed	γ^5 corresponds to the projection of the symmetry group of the regular dodecahedron	Chirality becomes a direction of geometrically irreversible symmetry breaking, ruling out right-hand coupling

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fermions (V-A structure)?	The left-hand sign is selected as stable by the regular Ricci curvature dynamics	
3. Charge Quantization and the Mystery of α Why is charge quantized? Why is $\alpha \approx 1/137$?	Charge = Chen class integral of a topological defect with symmetry breaking: $Q = \frac{1}{2\pi} \int \star dA$ Derived from the ground state of the ψ field: $\alpha = e^2 / (4\pi\epsilon_0 \hbar c)$	Theoretical value: $\alpha^{-1} = 137.036$ (deviation from the experimental value < 0.001%)
4. The Mystery of Leptonic Mixing Why do $\theta_{12}, \theta_{23}, \theta_{13}$, and Δm_{ij}^2 take specific values?	Closure of transformations under group operations $\sin\theta_{12} = \sqrt{\frac{3}{10}} \rightarrow \theta_{12} = 33.28^\circ$	$\Delta m_{21}^2 = 7.59 \times 10^{-5} \text{ eV}^2$ $\Delta m_{32}^2 = 2.51 \times 10^{-3} \text{ eV}^2$ (Consistent with data from JUNO 2025 and NuFIT 6.0)
	Golden Ratio $\tan\theta_{23} = \frac{\tan(\arctan(\phi)) - \tan\theta_{13}}{1 + \tan(\arctan(\phi))\tan\theta_{13}} \rightarrow \theta_{23} = 48.89^\circ$	
	Geometric Constraints $\sin\theta_{13} = \frac{1}{\phi^4} = 0.1459 \rightarrow \theta_{13} = 8.39^\circ$	

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	Derive the sum of squared deviations using the five-fold symmetric phase $t_{ij} = e^{2\pi i i-j /5}$	
5. The Mystery of Quark Mixing and CP Violation: Why Is the CKM Mixing Angle Small and the CP Phase $\delta_{CP} \approx 68.8^\circ$?	The H3 symmetry of the regular icosahedron $\rightarrow$ complex unitary geometry, related to $\phi = 1.61803$ The Bessel function at the dihedral angle $x = 2.2519$: $\tan \theta_{12} = j_2(x) = 0.2316 \rightarrow 13.04^\circ$	$J_{CP} = 3.04 \times 10^{-5}$ (PDG 2025 experimental value: $3.05 \pm 0.20 \times 10^{-5}$)
	$\tan \theta_{23} = j_2(x \approx 0.807) = 0.0413879 \rightarrow 2.40^\circ$	
	$\sin \theta_{13} = j_2(x \approx 0.234) \approx 0.003647 \rightarrow 0.210^\circ$	
	Biclinicity $\rightarrow \gamma = \pi/\phi^2 = 68.8^\circ$	

The closure of these five gaps is not achieved by introducing ad hoc parameters, but is naturally derived from the intrinsic rigidity of discrete geometry and topology. Those

seemingly arbitrary “magic numbers” in the Standard Model are, within the KnotSim framework, proven to be the “fingerprints” of the underlying discrete weaving structure as observed in macroscopic measurements.

The axioms of KnotSim are relatively concise: the underlying reality is a discrete network composed of dynamic connections. When these connections are sufficiently strong, they form stable topological structures—“knots” (Knot). Spacetime, matter, and forces all emerge from the weaving and resonance of “knots.” This paradigm shift—from “continuous symmetries” to “discrete topology”—enables us to interpret gauge groups, chirality, quantization rules, and even mixed angles as manifestations of the same geometrically rigid structure at different scales.

This book will systematically expound this theory, demonstrating how it unifies general relativity, quantum mechanics, and the Standard Model as emergent phenomena of a single discrete network. We invite readers to examine this cosmic blueprint woven from “knots” and witness how these once-open theoretical gaps close within the inevitability of geometry and topology.

This serves as the preface.
